Experiment 3: Visual-Inertial Odometry With Dynamic-Environment Filtering

Raw write-up for Chapter 4, Section 4.3

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Purpose of this document. This is the raw, complete write-up of Experiment 3, written before the thesis chapter; Section 4.3 of the final report is extracted from it. It holds both result slots, every table and figure, and the analysis. Every number traces to a file under `output/` named in the "Reporting rules" paragraph of its slot. Open concerns are set in red where they arise. The extreme-world campaign of September and October 2026 (scripted kinematic props, 66 repeats and 14 filtered runs) was retired on 6 November 2026 with its datasets and is not reported; its artefacts are under `trash/output/`.

4.3.1 Simulation

Status: partial. Unfiltered baselines only. Twenty runs on the four `dataset_allsensor_realistic` recordings (two layouts, each without and with physically simulated traffic): VINS-Fusion once per recording, ORB-SLAM3 with a CPU or GPU feature extractor and its loop closer active or disabled. Traffic is noted as a condition, not tested as a factor. No filtered run exists in the current dataset set, so the filtering hypothesis is untested, and the latency and resource measurement is taken from eighteen repeated runs on the two mapping recordings: closure-free ORB-SLAM3 with each front end, and VINS-Fusion, three repeats of every configuration.

Objective. Determine whether masking dynamic-object features improves localisation, at what compute cost, and whether tracking survives it. In the present state of the slot the question answered is the precondition: how each unfiltered pipeline behaves on the realistic warehouse, with and without traffic, and which of its own switches (front end, loop closer) decides its error.

Process. Every run replayed at rate 1.0 from the bag with the node live on the laptop, the deployment condition; no detector ran. VINS-Fusion ran once per recording (run `cpu`). ORB-SLAM3 ran four times per recording: the CPU or the GPU feature extractor (fork patches 10/10b, runs `*_cpu` and `*_gpu`), each with the loop closer active (`loopclosing_*`) or disabled (`noloopclosing_*`), all on 6 November 2026 with the same node binary and configuration (`config/wil_sim/stereo_imu.yaml`). One run per cell, so no within-condition spread exists.

Datasets. `realistic_000` (U layout) and `_001` (I layout) are the static twins; `_002` (U) and `_003` (I) carry ten and twelve moving props, pallet jacks on pick–carry–drop jobs and walking people. Routes are 103 to 171 m over 80 to 126 s (the dataset table of the Experiment 1 write-up); no two recordings share a route.

Reporting rules. Every run is reported on its own. Aligned ATE is RMSE after a rigid $SE(3)$ fit with scale fixed, in three dimensions; RPE at 1 s is translational over pairs that span no jump (a step above $10\,\mathrm{m\,s^{-1}}$ between consecutive poses). The loop-closer state is read from `loop_closing.csv` (sub-millisecond rows mean a disabled closer), not from the run name, and every closure the closer accepted is listed with its time and correction. Ground truth is the bag's `/ground_truth/odometry`, moved from the model origin to the IMU point both estimators report (lever arm 0.338 m forward, 0.192 m up) before alignment; the figures draw that moved path. "Kept" is the fraction of the bag's stereo pairs the node processed, so frames lost before the node's own counter count as lost. The onset analysis uses the start-anchored planar error against a threshold of three times the estimator's median error over its two static runs (VINS-Fusion 1.07 m, ORB-SLAM3 1.31 m). A filtered run, when one exists, is paired with the baseline of the same estimator, front end and loop-closer state, and the pair is admissible only under three gates: an active filter (`detection_match_rate` ≥ 0.98), processed frames and duration within 5 % of the baseline, and both runs on the same node binary. Sources: `output/compare/simulation/exp-realistic-ab/` (`runs_metrics.csv`, `baseline_table.tex`), `output/compare/simulation/exp3-timelines/` (`timelines.csv`, `thresholds.csv`), run directories under `output/output_orb/simulation/` and `output/output_vins/simulation/`. The run that first occupied `realistic_002/loopclosing_gpu` had been launched on the `realistic_003` bag with the wrong output path (its manifest names that bag and world, and it received 3 535 stereo pairs where `realistic_002` holds 2 409); it is set aside under `trash/output/misfiled_20261106/` and the cell was re-run on the right bag on the evening of 6 November.

Results. VINS-Fusion holds 0.22–0.66 m aligned ATE on the four recordings with no jump and 1.2–3.6° of rotation error; its worst recording is the U layout with traffic, 0.66 m against 0.29 m on that layout's static twin, while on the I layout traffic changes nothing (0.27 against 0.22 m). ORB-SLAM3 spans 0.14–6.02 m: eleven of its sixteen runs lie between 0.14 and 0.52 m and five diverge, and which five is decided by the loop closer and the recording, not by the front end (table 3.1, figures 3.1 and 3.2). Neither front end loses tracking on any recording, with or without traffic: zero tracking-loss events and zero resets in twenty runs.

Two columns of table 3.1 qualify those RMSE figures. The first is the worst single error each run reaches after its initialisation transient, which the RMSE averages away: over the fifteen runs that held it is 0.26–1.25 m, between 0.8 and 2.6 times the run's own RMSE, so no run that tracked was ever more than about 1.25 m from the truth; over the five that diverged it is 2.25–13.42 m. The two orderings are not the same one. On `realistic_000` VINS-Fusion has the lower RMSE (0.290 against 0.338 m) and ORB-SLAM3 the smaller worst case (0.260 against 0.576 m), so which estimator is "better" on that recording depends on whether the question is average accuracy or the largest excursion — and for a vehicle in a 3.0 m aisle it is the second. The exclusion matters: every ORB-SLAM3 run peaks near 2.5 s at three to four metres on the gravity re-alignment that ends its inertial initialisation, and a raw maximum would report that initialisation as though it were driving.

The second column is the same ATE as a percentage of the ground-truth path length. It exists so that these 103–172 m routes can be set beside the real robot's much shorter ones, and it separates the two outcomes more sharply than the metre figure does: 0.09–0.64 % for the fifteen runs that held against 1.11–4.51 % for the five that diverged. It is a ratio of two quantities already in the table, not a KITTI-style sub-sequence drift rate, and it is not comparable to one.

Analysis: the loop closer decides, in both directions. On the static I layout both loop-closing runs accept one closure at 73.2 s, in the heading change at 76 % of the route, with a correction of 14.7 m (CPU) and 13.3 m (GPU) and 1.9–2.8°; a global bundle adjustment follows and the trajectory is 6.02 and 4.78 m off for the rest of the run, where the two closure-free runs on the same recording hold 0.50 and 0.40 m (figure 3.4). A route with half a metre of accumulated error cannot need a fourteen-metre correction, so the match is false: the I layout's long parallel aisles present the same rack faces twice, and the matched

Table 3.1: Unfiltered arms on the four realistic-warehouse recordings, 6 November 2026, replay rate 1.0. Aligned ATE is RMSE after a rigid $SE(3)$ fit with scale fixed; RPE is translational, at $\Delta = 1$ s over pairs that do not span a jump; "Jumps" counts steps above $10\,\mathrm{m\,s^{-1}}$; "Kept" is the fraction of the bag's stereo pairs the estimator processed (frames lost before the node's own queue count as lost). Loop-closer state is read from `loop_closing.csv`, not from the run name. Static and dynamic rows of one layout share a world geometry and differ in traffic only. "Max" is the largest single position error the run reaches *after* its initialisation transient — the same cut the onset analysis of table 3.2 makes, after the last jump inside the first 15 s — on the same alignment as the ATE column, and records the excursion that the RMSE averages away. Every ORB-SLAM3 run peaks near 2.5 s on the gravity re-alignment that ends its inertial initialisation, at three to four metres; that peak is excluded here because it measures the initialisation and not the driving. "ATE/path" is the aligned ATE as a percentage of the ground-truth path length, so that runs over routes of different length can be compared; it is a ratio, not a sub-sequence drift rate. **Bold** marks the lowest ATE, the lowest Max and the lowest jump-free RPE among the five arms of a recording; with one run per arm it records which run came lowest, not that the arm is better.

Recording	Layout	Scene	Estimator	ATE [m]	Max [m]	ATE/path [%]	Rot. [°]	$\mathrm{RPE}_{1\,\mathrm{s}}$ [m]	Jumps	Kept	Resets
`realistic_000`	U	static	ORB-SLAM3	0.338	**0.260**	0.20	13.74	0.083	2	0.72	0
			ORB-SLAM3 GPU	0.293	0.525	0.17	12.80	0.079	3	0.99	0
			ORB-SLAM3 (no LC)	0.483	1.249	0.28	14.05	0.090	2	0.74	0
			ORB-SLAM3 GPU (no LC)	0.523	0.995	0.30	12.84	**0.078**	3	0.99	0
			VINS-Fusion	**0.290**	0.576	0.17	1.85	0.084	0	0.77	0
`realistic_001`	I	static	ORB-SLAM3	6.021	12.591	4.51	16.92	0.094	6	0.73	0
			ORB-SLAM3 GPU	4.780	13.421	3.58	18.98	0.092	7	0.99	0
			ORB-SLAM3 (no LC)	0.496	0.642	0.37	16.05	0.093	1	0.74	0
			ORB-SLAM3 GPU (no LC)	0.400	0.514	0.30	15.13	**0.088**	2	0.97	0
			VINS-Fusion	**0.221**	**0.456**	0.17	1.18	0.089	0	0.78	0
`realistic_002`	U	dynamic	ORB-SLAM3	1.393	2.254	1.35	16.99	**0.078**	4	0.77	0
			ORB-SLAM3 GPU	0.400	0.595	0.39	16.04	**0.078**	4	0.98	0
			ORB-SLAM3 (no LC)	0.293	0.501	0.28	17.12	0.080	2	0.83	0
			ORB-SLAM3 GPU (no LC)	**0.170**	**0.255**	0.16	16.08	0.083	1	0.99	0
			VINS-Fusion	0.663	1.136	0.64	3.57	0.116	0	0.88	0
`realistic_003`	I	dynamic	ORB-SLAM3	**0.139**	**0.316**	0.09	13.52	0.075	2	0.94	0
			ORB-SLAM3 GPU	0.504	0.756	0.32	13.16	0.076	3	0.99	0
			ORB-SLAM3 (no LC)	1.755	2.741	1.11	14.22	0.080	2	0.88	0
			ORB-SLAM3 GPU (no LC)	1.911	2.800	1.21	13.53	0.083	3	0.99	0
			VINS-Fusion	0.268	0.537	0.17	1.51	**0.071**	0	0.87	0

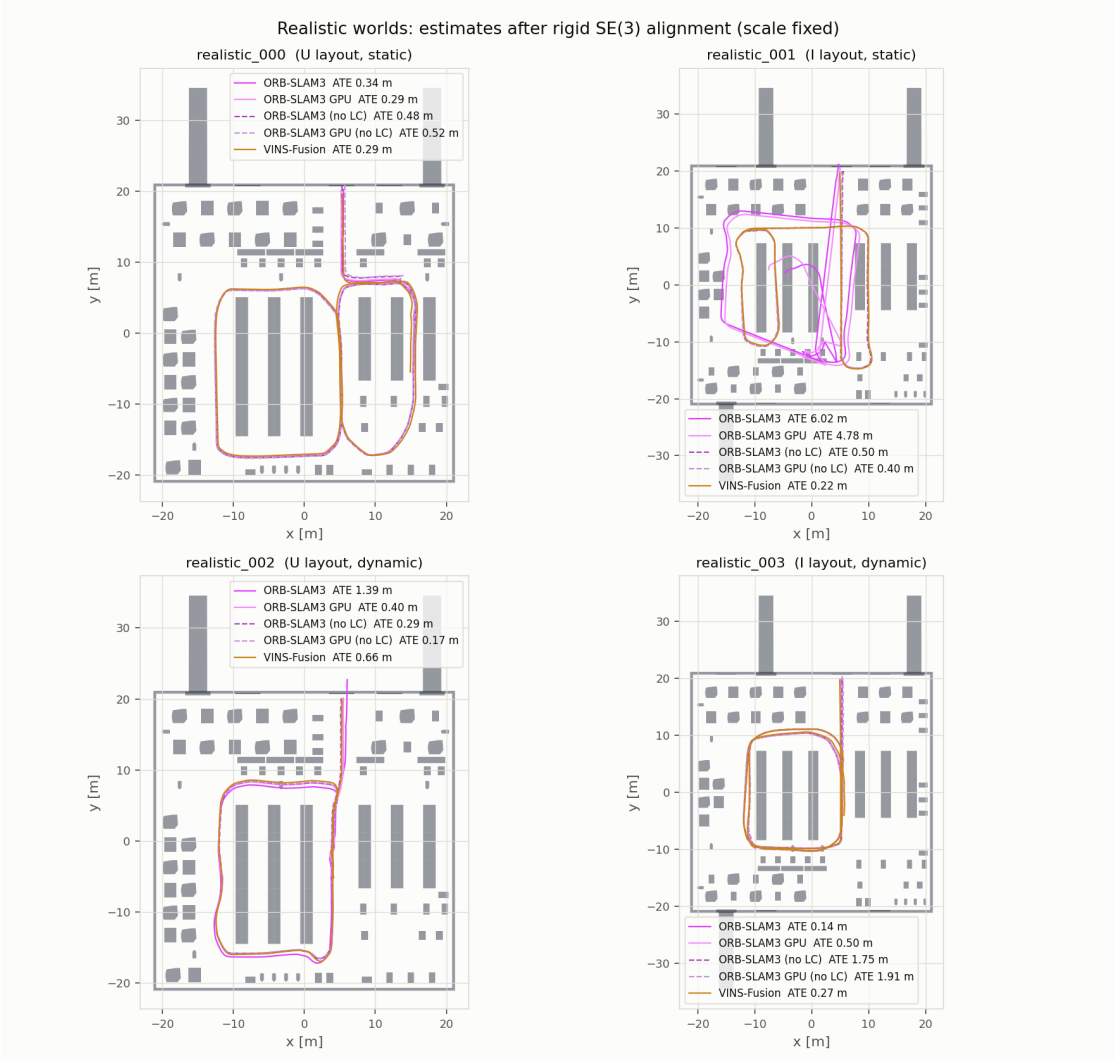


Figure 3.1: Unfiltered estimates on the four realistic-warehouse recordings after rigid $SE(3)$ alignment with scale fixed, on the warehouse footprint. Top: the static twins; bottom: the same layouts with traffic. Solid magenta is ORB-SLAM3 with its loop closer active, dashed with it disabled, lighter for the GPU front end; orange is VINS-Fusion. The two excursions on `realistic_001` are the loop-closing runs after their false closure; the 1.4 m run on `realistic_002` is the CPU loop-closing run; the closure-free runs on `realistic_003` lie on the route in plan and diverge vertically.

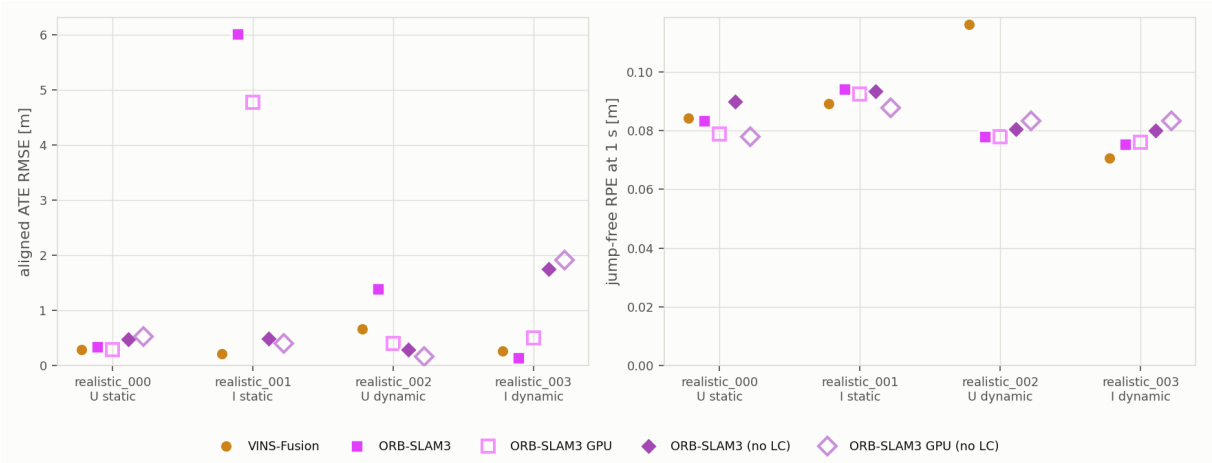


Figure 3.2: Aligned ATE (left) and jump-free translational RPE at 1 s (right) of every unfiltered run, one marker per run; filled markers are the CPU front end, open markers the GPU front end of the same configuration. The relative error is 0.07–0.12 m for every run, so the ATE differences are discrete events, not tracking quality.

keyframes (97 against 255, 104 against 272) are one aisle apart. The same candidate was then re-proposed four times and rejected by the inertial geometry check, after the damage. That both front ends accept it at the same frame with the same correction says the place-recognition stage, not the extractor, is where the fault lies.

On the I layout with traffic the sign reverses. Both loop-closing runs accept one closure at 56.5 s (48 % of the route) with a 0.33 m correction and hold 0.14 and 0.50 m; the closure-free runs read 1.76 and 1.91 m, yet their planar error never exceeds 0.7 m (figure 3.6). The difference is vertical: the closure-free estimates climb to 4.7 and 5.1 m above the floor by the end of the run while the loop-closing runs stay within 0.6 and 1.4 m, and the three-dimensional ATE reports that climb. The GPU loop-closing run on this recording also rejected 148 further candidates by inertial geometry, every one of them after its single accepted closure. On the U layout the closer is nearly inert: two accepted closures per run on `realistic_000` with corrections of 0.07–0.12 m, one of 0.26–0.33 m per run on `realistic_002`, and the eight U-layout ORB-SLAM3 runs within 0.17–0.52 m except the CPU loop-closing run on `realistic_002`, whose 1.39 m is an offset present from 6.5 s on, before any closure (figure 3.5); with four jumps and the lowest relative error on the recording it is an initialisation event, not a tracking one, and the GPU run of the same configuration, which accepts the same closure at 65 s, does not reproduce it (0.40 m).

Analysis: the front end changes throughput, not accuracy. The GPU extractor processes 97–99 % of each bag's frames where the CPU extractor processes 72–94 %, and its tracking median is 30–37 ms against 49–55 ms (from the per-run `performance.csv`). Across the eight CPU/GPU pairs with the same closer state the ATE differs by at most 0.16 m in five; the three larger differences (−1.2, −1.0 and 0.4 m) are each one run's discrete event, in both directions, and the two divergence patterns above occur in both front ends alike. At this frame budget the extractor is not the limiting stage of accuracy on these recordings.

Analysis: traffic as a condition. VINS-Fusion is 0.37 m worse on the U layout with traffic than on its static twin and 0.05 m worse on the I layout; its relative error on `realistic_002` is the highest of any run (0.116 m) and its error trace carries two waves of 0.7 m at 25 and 45 s on the western and southern aisles, where the pallet jacks work, that no static run shows (figure 3.5). ORB-SLAM3's closure-free runs are *better* on the U traffic recording (0.29, 0.17 m) than on its static twin (0.48, 0.52 m). With one run per cell and routes that differ between twin and traffic recording, neither number is a traffic effect; what the slot establishes is that physically simulated traffic degrades the tracker that follows features longest, and breaks neither.

Analysis: the rotation column. ORB-SLAM3's 13–19° rotational RMSE is one event on every run: the first jump lies at 2.5–4.8 s (table 3.2), where inertial initialisation re-aligns the world frame to gravity, and after it the clean rotational RPE is 0.07–0.33°. The jump detector is translational, so the column includes the event; the median is the number to read.

Table 3.2: Realistic world, unfiltered runs: onset of sustained error and its place on the route, per run. Onset is the first time the start-anchored position error stays above 3× the estimator's median error over its static runs (VINS-Fusion 1.07 m, ORB-SLAM3 1.31 m) for the rest of the run; "where" is the route segment and the share of the route completed at that time; the first jump is a step above 10 m s^{-1} between consecutive poses. "never" means the error stayed below the threshold for the whole run.

Recording	Estimator	LC	Onset [s] (where)	First jump [s]
`realistic_000`	ORB-SLAM3	active	never	4
	ORB-SLAM3	disabled	never	4
	ORB-SLAM3 GPU	active	never	4
	ORB-SLAM3 GPU	disabled	60 (straight 48 %)	4
	VINS-Fusion	—	never	—
`realistic_001`	ORB-SLAM3	active	73 (turn 78 %)	5
	ORB-SLAM3	disabled	never	5
	ORB-SLAM3 GPU	active	73 (turn 78 %)	5
	ORB-SLAM3 GPU	disabled	never	5
	VINS-Fusion	—	never	—
`realistic_002`	ORB-SLAM3	active	6 (straight 6 %)	3
	ORB-SLAM3	disabled	never	3
	ORB-SLAM3 GPU	active	never	4
	ORB-SLAM3 GPU	disabled	never	3
	VINS-Fusion	—	never	—
`realistic_003`	ORB-SLAM3	active	never	3
	ORB-SLAM3	disabled	never	3
	ORB-SLAM3 GPU	active	never	3
	ORB-SLAM3 GPU	disabled	never	3
	VINS-Fusion	—	never	—

Filtered arms. untested. The current dataset set holds no filtered run: the October 2026 filtered runs on these recordings were retired on 6 November 2026 with the rest of the earlier campaign, and the filter has not been re-run under the present pipeline (GPU front end, patch-15 initialisation switches). When it is, each filtered run pairs with the baseline of the same estimator, front end and closer state under the three gates above, and the loop-closer finding here fixes the pairing that matters: on the I layout the comparison must be made with the closer disabled, or the false closure will dominate both arms.

Latency and resources. By the design decision of 6 November 2026 the real-time and resource measurement of this experiment is taken only on the two mapping recordings, each configuration run three times: `dataset_allsensor_realistic_map` (U layout, 286 s, 8 725 stereo pairs) and `dataset_allsensor_realistic_01_map` (I layout, 315 s, 9 552 pairs). Eighteen runs exist, six configurations of three repeats each at replay rate 1.0 with no detector: ORB-SLAM3 with the CPU and with the GPU front end, loop closer disabled, and VINS-Fusion (table 3.3). The repeats are tight, front-end medians within 1–7 ms of their mean and peak memory within 30 Mi B.

The two pipelines divide the cost differently, and neither is cheaper than the other outright. VINS-Fusion optimises in 44 ms where ORB-SLAM3's local mapping takes 211–406 ms, and holds its state in

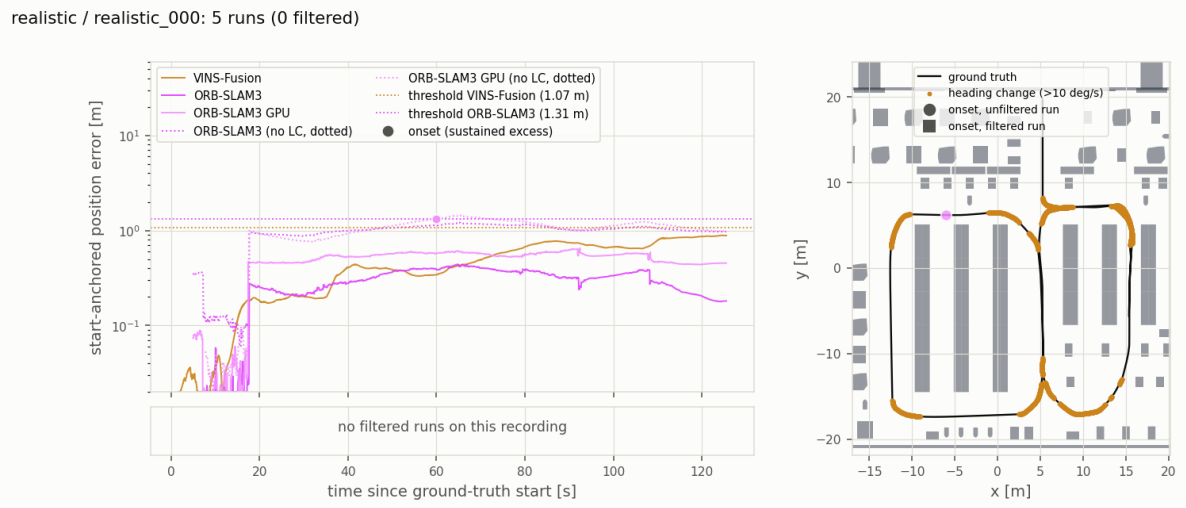


Figure 3.3: `realistic_000` (U, static), every run. Left: start-anchored planar error on a log axis, solid for loop closing active, dotted for disabled, lighter for the GPU front end, with each estimator's threshold; the dot marks the onset of sustained excess. Right: the route with heading changes in orange and every onset placed on it. The one onset is the GPU closure-free run at 60 s, a sustained 1.3 m offset on the straight, not a jump.

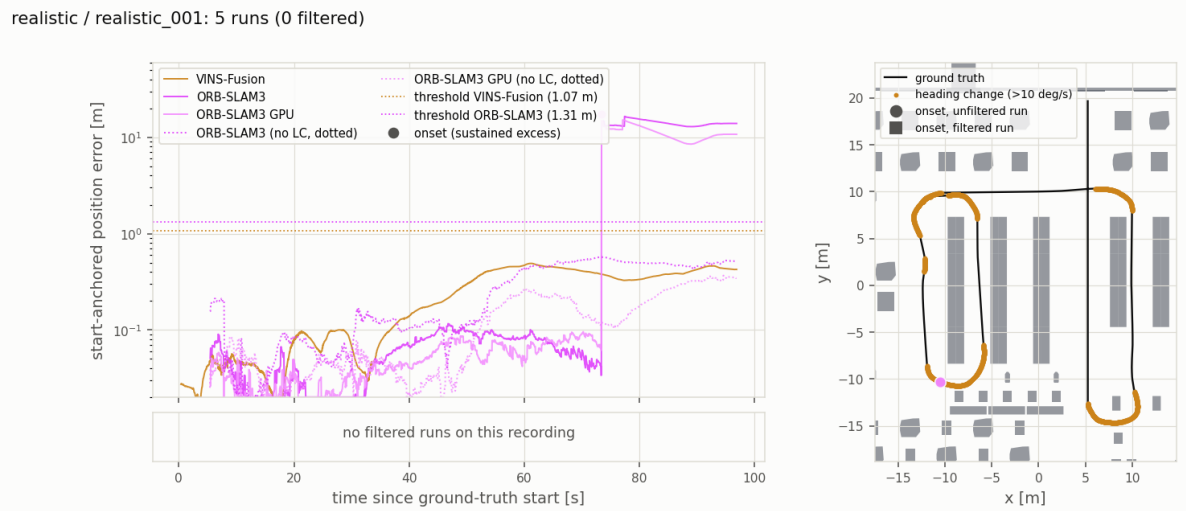


Figure 3.4: `realistic_001` (I, static), every run; panels as in figure 3.3. Both loop-closing runs step to 14 m of error at 73 s, in the heading change at 76 % of the route, at the single accepted closure; the closure-free runs and VINS-Fusion never cross.



Figure 3.5: `realistic_002` (U, traffic), every run; panels as in figure 3.3. The CPU loop-closing run carries a 1.5 m offset from 6.5 s, which the GPU loop-closing run does not share; VINS-Fusion's two 0.7 m waves at 25 and 45 s lie on the aisles where the pallet jacks work.

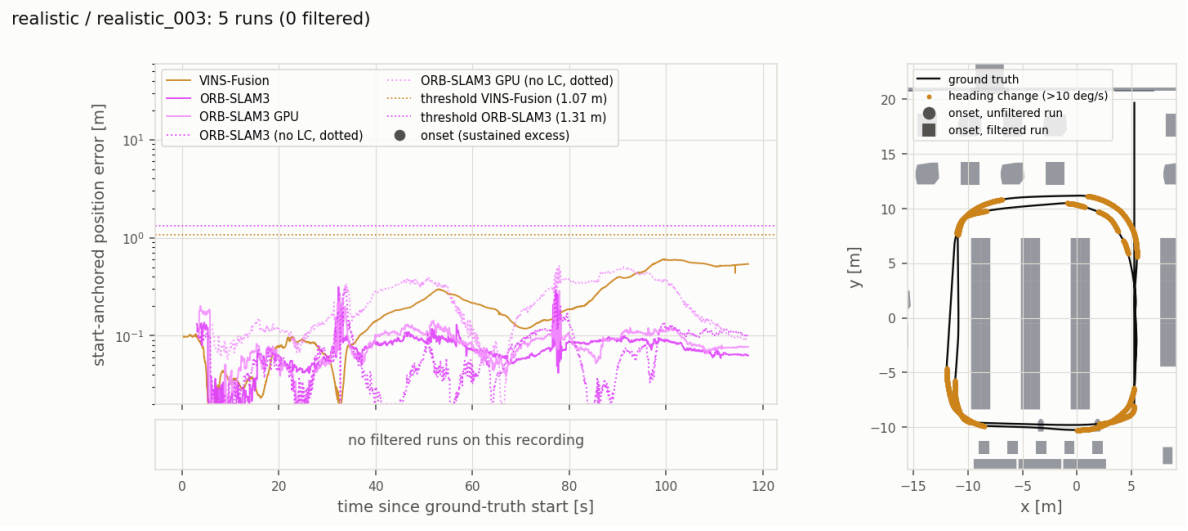


Figure 3.6: `realistic_003` (I, traffic), every run; panels as in figure 3.3. No run crosses its threshold in the plane; the 1.8 and 1.9 m ATE of the closure-free runs is vertical drift, which the single accepted closure of the loop-closing runs removes.

a quarter of the memory, but spends more than twice the processor time: about eight cores of one against three and a half. Their front ends are within 6 ms of each other, and every configuration keeps 75–96 % of the bag's frames, so the choice between them is a memory-against-cores trade, not a speed one.

Between ORB-SLAM3's two front ends the recordings disagree. On the U recording the GPU extractor behaves as the single runs showed, tracking in 37 against 42 ms and keeping 96 against 93 % of frames; on the I recording the sign reverses, 52 against 44 ms and 75 against 91 % kept, with the local-mapping median at 406 ms. VINS-Fusion reads the same on both recordings (front end 38.1 and 38.6, back end 44.3 and 44.5 ms), so the recordings themselves do not impose a difference of that size.

[Confound: run order] The six GPU runs were made consecutively from 21:04 to 22:04 and the six CPU runs from 23:27 to 00:18, so the I-recording GPU repeats were the fourth to sixth GPU runs of a session on a laptop that throttles at 90 °C; the front-end record notes that timing comparisons need a cool-start, interleaved order. The six VINS-Fusion runs are a separate session of 7 November and are not a control for that state. The U-recording figures are the ones to quote until the I recording is repeated interleaved.

Table 3.3: Real-time and resource measurements on the two mapping recordings, replay rate 1.0, no detector: mean ± SD over n repeats of each configuration. **Bold** marks the best mean in each column within a recording. Front-end and back-end are each run's median module latency (ORB-SLAM3: tracking and local mapping; VINS-Fusion: feature tracking and back-end optimisation); throughput is processed frames per second of sensor time; "Kept" is the processed fraction of the bag's stereo pairs; CPU is process time over wall time, in percent of one core; RSS the peak resident set. The loop-closer state is read from `loop_closing.csv`.

Recording	Estimator	LC	n	Front-end [ms]	Back-end [ms]	Throughput [Hz]	Kept [%]	CPU [%]	Peak RSS [Mi B]
`realistic_map`	ORB-SLAM3	disabled	3	42.0 ± 3.3	211.3 ± 14.8	28.3 ± 0.9	93 ± 3	379 ± 11	1732 ± 15
	ORB-SLAM3 GPU	disabled	3	**36.7 ± 1.2**	284.2 ± 14.7	**29.1 ± 0.4**	**96 ± 1**	**332 ± 7**	1794 ± 19
	VINS-Fusion	—	3	38.1 ± 3.3	**44.3 ± 5.0**	27.5 ± 2.7	91 ± 9	819 ± 10	**479 ± 15**
`realistic_01_map`	ORB-SLAM3	disabled	3	43.9 ± 1.0	220.4 ± 12.6	**27.5 ± 0.3**	**91 ± 1**	381 ± 6	1816 ± 29
	ORB-SLAM3 GPU	disabled	3	51.9 ± 3.8	405.7 ± 22.6	22.9 ± 1.4	75 ± 4	**361 ± 9**	1719 ± 21
	VINS-Fusion	—	3	**38.6 ± 0.8**	**44.5 ± 1.5**	27.4 ± 0.5	90 ± 2	814 ± 5	**497 ± 11**

What each accuracy run cost. The twenty accuracy runs were measured for cost as well, on the same machine and in the same sessions, and table 3.4 reports them one row per run. They are not a substitute for table 3.3: they are single runs on four short recordings rather than three repeats on two long ones, and the divergences of table 3.1 are inside them. What they add is that accuracy and cost can be read on the same runs, which figure 3.7 draws.

The separation between the two pipelines is the one the mapping recordings show, at the same sizes. VINS-Fusion holds its state in 477–492 Mi B against ORB-SLAM3's 1029–1350 Mi B and optimises in 45–60 ms against 189–286 ms, for 707–801 % of one core against 285–448 %. Within ORB-SLAM3 the GPU front end tracks in 29.6–36.7 ms against the CPU front end's 49.2–54.9 ms and draws about a core less, at 150 Mi B more memory. These runs were not interleaved either, so they carry the same thermal caveat as table 3.3 and are quoted as a second observation of the same separation, not as an independent one.

The figure's own result is a negative one: no panel shows cost buying accuracy. Every run that diverged is ORB-SLAM3, and within each front end the diverged runs cost what the runs that held cost — with the GPU front end 31.6–36.7 ms against 29.6–36.2 ms, overlapping entirely, and with the CPU front end 52.6–54.9 ms against 49.2–51.6 ms, a gap of one to five milliseconds across three runs and two recordings, which one run per cell cannot separate from the recording. A run about to be thrown several metres by a false loop closure is not working harder beforehand, so no resource measure on this machine is a usable early warning for the failure mode this experiment actually found.

How repeatable is any of this? The eighteen runs above are the only repeated runs in the simulation campaign, and scoring them for accuracy as well as for time measures something no other result in this chapter can: how much an estimate moves when nothing changes. Table 3.5 gives the spread over the three

Table 3.4: Realistic world, real-time and resource measurements, one row per run. Every run is the deployment condition of section 4.3.1: replay rate 1.0, node live on the laptop, no detector running, which is why the mask column reads *off* throughout. Front-end and back-end are each run's median module latency (ORB-SLAM3: tracking and local mapping; VINS-Fusion: feature tracking and back-end optimisation); throughput is processed frames per second of sensor time; "Kept" is the processed fraction of the bag's frames, so frames lost before the node's own queue count as lost; CPU is process time over wall time, in percent of one core, so a value above 100 is more than one core; RSS the peak resident set. Accuracy for the same runs is in table 3.1.

Recording	Estimator	Mask	LC	Atlas	Front-end [ms]	Back-end [ms]	Throughput [Hz]	Kept	CPU [%]	Peak RSS [Mi B]
realistic_000	ORB-SLAM3	off	active	no	51.6	272.0	21.7	72 %	445	1133
	ORB-SLAM3	off	disabled	no	50.4	258.0	22.4	74 %	438	1153
	ORB-SLAM3 GPU	off	active	no	29.6	226.8	29.9	99 %	340	1350
	ORB-SLAM3 GPU	off	disabled	no	31.4	251.7	29.9	99 %	350	1347
	VINS-Fusion	off	—	no	39.0	44.9	23.5	77 %	801	477
realistic_001	ORB-SLAM3	off	active	no	52.6	274.5	22.3	73 %	448	1162
	ORB-SLAM3	off	disabled	no	51.5	271.3	22.6	74 %	437	1057
	ORB-SLAM3 GPU	off	active	no	31.6	233.2	29.9	99 %	343	1349
	ORB-SLAM3 GPU	off	disabled	no	34.5	271.2	29.3	97 %	353	1264
	VINS-Fusion	off	—	no	40.8	49.9	23.8	78 %	730	487
realistic_002	ORB-SLAM3	off	active	no	54.9	285.8	23.4	77 %	439	1135
	ORB-SLAM3	off	disabled	no	50.8	262.4	25.4	83 %	428	1029
	ORB-SLAM3 GPU	off	active	no	30.7	189.1	30.0	98 %	285	1334
	ORB-SLAM3 GPU	off	disabled	no	34.6	262.6	29.9	99 %	327	1215
	VINS-Fusion	off	—	no	40.5	49.5	26.7	88 %	738	482
realistic_003	ORB-SLAM3	off	active	no	49.2	267.3	28.5	94 %	408	1141
	ORB-SLAM3	off	disabled	no	54.1	279.3	26.6	88 %	423	1157
	ORB-SLAM3 GPU	off	active	no	36.2	250.3	30.1	99 %	306	1346
	ORB-SLAM3 GPU	off	disabled	no	36.7	269.6	30.0	99 %	309	1335
	VINS-Fusion	off	—	no	46.0	60.3	26.3	87 %	707	492

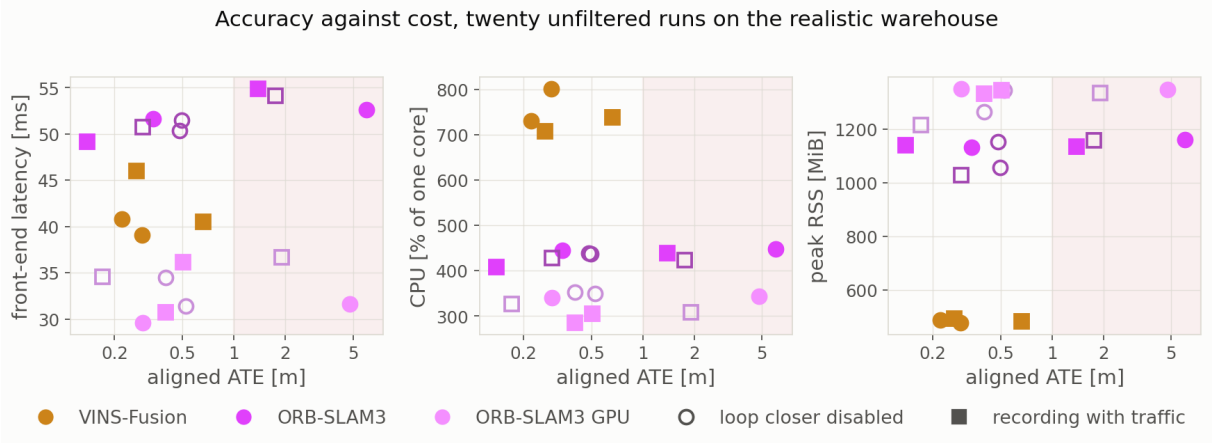


Figure 3.7: Accuracy against cost, one point per unfiltered run on the four realistic-warehouse recordings, from tables 3.1 and 3.4. Aligned ATE is on a logarithmic axis in all three panels; the shaded band is the region above 1 m, which holds the five diverged runs. Orange is VINS-Fusion, magenta ORB-SLAM3 with the CPU front end and light magenta with the GPU front end; an open marker is a run with the loop closer disabled and a square a recording carrying traffic. Cost is the run's median front-end latency, its process time as a percentage of one core, and its peak resident set.

repeats of each configuration. The two estimators answer differently. VINS-Fusion repeats to 0.094 m on the U recording and 0.035 on the I, five and seven per cent of its own mean. ORB-SLAM3 does not: its three repeats span 0.089 m on the U recording and 0.277 on the I with the CPU front end, and 0.016 and 0.236 m with the GPU front end — on the I recording that is half to four fifths of the configuration's own mean, from the same binary, the same configuration file and the same bag, with the loop closer disabled in all six.

The frame counts in the same table are consistent with why. VINS-Fusion processed every frame of both recordings in all six runs, so it saw identical input each time; ORB-SLAM3 kept 90–97 % with the CPU front end and 70–98 % with the GPU one, and which frames a run loses depends on how the laptop was scheduling at that moment. That is a consistent explanation and not a demonstrated one: a keyframe-based system with a bundle adjustment on its own thread need not be deterministic even on identical input. Separating the two would need a run at a replay rate slow enough to drop nothing, which this chapter does not have. The worst spread also belongs to the I recording, whose repeats carry the thermal confound noted above, so part of 0.277 m may be the laptop rather than the estimator.

Read against table 3.1 this sets a floor under what that table can be asked. A difference of less than about a quarter of a metre between two ORB-SLAM3 arms is inside the spread measured here and should not be read as an effect; this is the quantity the bold marking of that table was hedged against, now measured. Two of the chapter's findings survive it and one is strengthened. The five divergences are 1.39–6.02 m, five to twenty times the floor, and the false closure that causes the largest of them is reproduced by both front ends on the same frame. The claim that the front end changes throughput and not accuracy is supported rather than weakened: the CPU and GPU arms differ by less than this floor on every recording, which is what having no effect looks like. What should not be carried further is any ordering of two ORB-SLAM3 arms that rests on a gap smaller than it.

Table 3.5: Run-to-run spread over the three repeats of each configuration on the two mapping recordings, the only repeated runs in the simulation campaign; the same eighteen runs as table 3.3, scored for accuracy. Every run has its loop closer disabled, read from `loop_closing.csv`. "ATE" is mean ± SD over the three repeats and "spread" the largest difference between two of them; "Max" is the largest error after the initialisation transient, on the convention of table 3.1. The three repeats of a configuration were run back to back rather than interleaved, so a drift across them is confounded with the laptop's thermal state (section 4.3.1); the Kept column is where that would show.

Recording	Estimator	n	ATE [m]	spread [m]	Max [m]	ATE/path [%]	Kept
`realistic_map`	ORB-SLAM3	3	0.189 ± 0.045	0.089	0.748 ± 0.267	0.05 ± 0.01	0.94 ± 0.03
	ORB-SLAM3 GPU	3	0.144 ± 0.008	0.016	0.345 ± 0.033	0.04 ± 0.00	0.96 ± 0.01
	VINS-Fusion	3	1.915 ± 0.049	0.094	3.617 ± 0.239	0.51 ± 0.01	1.00 ± 0.00
`realistic_01_map`	ORB-SLAM3	3	0.348 ± 0.149	0.277	1.517 ± 0.749	0.09 ± 0.04	0.91 ± 0.01
	ORB-SLAM3 GPU	3	0.433 ± 0.118	0.236	2.691 ± 0.338	0.11 ± 0.03	0.76 ± 0.04
	VINS-Fusion	3	0.491 ± 0.017	0.035	5.088 ± 0.167	0.13 ± 0.00	1.00 ± 0.00

Limitations, each stated once. *n:* one run per cell on the four accuracy recordings; the three repeats of each mapping-recording configuration (table 3.5) are the only within-configuration spread the campaign measures, and they are on different routes. *Route and traffic confounded:* no two recordings share a route. *Three-dimensional ATE:* the metric charges vertical drift to a floor robot; the planar onset table is the complement, not a replacement. *Deployment rate on one laptop:* frame retention and every latency figure are properties of this machine at rate 1.0, and the mapping-bag repeats were not interleaved between front ends.

[Open: the false closure is diagnosed from its correction, not from the images] The 14 m correction and the one-aisle keyframe separation identify the `realistic_001` closure as false, and both front ends agree; the matched image pair has not been inspected, and the two April-style guards that would catch it (a covisibility-consistency window, or a maximum correction against the inertial drift bound) are

untested. Until one is, the finding is a diagnosis, not a fix.

Summary. On the realistic warehouse neither pipeline loses tracking, with or without traffic. VINS-Fusion holds 0.22–0.66 m with no jump and is worst on the U layout with traffic; ORB-SLAM3 holds 0.14–0.52 m in eleven runs of sixteen, and its five divergences are the loop closer's: a false closure on the I layout's repeated aisles that puts both front ends 4.8–6.0 m off, a vertical drift on the same layout with traffic that only the closer corrects, and one initialisation offset. The front end changes throughput (97–99 % of frames kept against 72–94 %) and not accuracy. The filtering hypothesis is untested in the current dataset set. On the mapping recordings VINS-Fusion optimises five to nine times faster than ORB-SLAM3's local mapping and in a quarter of the memory, for more than twice the processor time; within ORB-SLAM3 the GPU front end tracks in 37 ms and keeps 96 % of frames against the CPU front end's 42 ms and 93 % on the U layout, in three repeats each, and the I-layout repeats contradict that and were run in a thermally confounded order.

Future plan. Run both pipelines filtered on `realistic_002` and `_003` with the detector live, paired under the three gates and with the closer disabled on the I layout; add a correction bound to the loop closer and test it on `realistic_001`; repeat the mapping-recording runs interleaved between all three configurations from a cool start, so that the thermal state is shared rather than confounded with the configuration; and record a second route in each layout so that traffic and route are separable.

4.3.2 Real robot

Status: partial. Four real recordings with an independent planar reference. The first, `dataset_realv6_000` (31 October 2026), carries five unfiltered runs whose raw data exist: ORB-SLAM3 stereo-inertial with GPU or CPU front end and loop closing on or off, and VINS-Fusion once. ORB-SLAM3 tracks the whole route in all four configurations at a provisional 0.46 to 1.03 m planar ATE, the GPU front end at 0.46–0.52 m; VINS-Fusion resets 217 times on IMU-stream gaps and has no valid run. No filtered run exists, the `realv6` tracker export fails the pre-registered validity rules, the three `realv7` exports pass them and all twelve ORB-SLAM3 runs on them score after a frame repair in the scorer (0.25–0.27 m on `000`), but the IMU stayed uninitialised for most of nine runs (an upstream map reset at the initialisation, then a starved re-initialisation). On 11 November all twelve were re-made with the fork's patch-15 switches active: the upstream reset does not occur in any of them, every estimate is level, and scored with no repair at all they give 0.251–0.268 m on `000`, 0.442–1.278 on `_001` and 0.172–0.198 on `_002`, seven of the twelve **completed** where none was before. Patch 15 is therefore exercised, and what it establishes is that the scorer's repair is no longer needed, not that the estimator is more accurate: against the repaired stock figures it matches on `000`, is mixed on `_001` and wins on `_002`. VINS-Fusion has no valid run, the marker-to-IMU lever arm is unmeasured, and no filtered run exists, so the filter hypothesis remains untested on the real robot and no headline is claimed.

Objective. Establish whether the two unfiltered pipelines track the real robot against an independent reference, as the precondition for a filtered-versus-unfiltered test on real imagery.

Process. The bag was replayed into each node at real time on the afternoon of 3 November 2026; no detector ran. ORB-SLAM3 used `config/wil/stereo_imu.yaml` with the repaired library (fork patches 11–14b, node and library built alike), the 3 November stereo extrinsic and `IMU.NoiseGyro` 0.1 with `IMU.NoiseAcc` 1.0, ten times the values that had been copied from the VINS file, in four configurations: GPU or CPU feature extractor, loop closing on or off. VINS-Fusion used the same file with the gravity norm of an August recording. Earlier runs on this bag were overwritten by these and are not reported. The optical tracker export was scored with the real-robot procedure of Experiment 1.

Dataset. 220 s, 6 601 stereo pairs at 30 Hz, RealSense IMU at 200 Hz on the inherited topic name, wheel-speed feedback, a figure-of-eight of about 27 m at a median 0.15 m s^{-1} on the Gensurv floor, no scripted moving objects. Tracker export: 277 samples, median spacing 0.6 s, 66 gaps above 1 s, one sample 5.2 m off the path at 184.9 s.

Reporting rules. `script/eval_real_gt.py`: rigid SE(2) alignment with scale fixed, lever arm set to zero and unmeasured, every value provisional. The export fails the tool's gap and step rules, so the tool labels each run *failed (ground truth invalid)*; the values below are diagnostics. All are scored with the one off-path sample removed, an edit recorded here and applied nowhere else; table 3.7 shows what that sample does to one run. The bag was not on the machine, so the time offset is pinned to the value the wheel-speed cross-correlation found on 2 November and the manifests record the substitution. Sources: `output/compare/real/realv6_1103pm_*/`, `output/compare/real/realv6-insights/`, run directories under `output/output_orb/real/dataset_realv6_000/` and `output/output_vins/real/dataset_realv6_000/`.

Results. ORB-SLAM3 follows the whole 218 s route in every configuration: 0.52 and 0.46 m with the GPU extractor (loop closing on and off), 0.62 and 1.03 m with the CPU extractor, which drops two thirds of the frames at real time (table 3.6, figure 3.8). VINS-Fusion tracked 18 s before 217 IMU-gap resets took over (table 3.9) and has no entry in the accuracy table.

Table 3.6: ORB-SLAM3 on `dataset_realv6_000` against the cleaned tracker truth, 3 November 2026 afternoon series: repaired library, replay rate 1.0, 3 November stereo extrinsic, `IMU.NoiseGyro` 0.1 and `IMU.NoiseAcc` 1.0, GPU or CPU feature extractor, loop closing (LC) on or off. Aligned planar ATE after a rigid SE(2) fit with scale fixed; RPE at 1 s translational; path ratio = estimate / truth; resets = `local_mapping_bad_imu` map resets; dropped = camera frames the node's queue discarded of about 6 480 received. Lever arm tracker-marker to IMU unmeasured (zero applied); time offset pinned to −4.14 s; all values provisional under the scorer's validity rules.
textbfBold marks the lowest ATE of the five runs; with one run per configuration that records which run came lowest, not that the configuration is better.

Run	ATE [m]	median [m]	max [m]	$RPE_{1\,s}$ [m]	path ratio	resets	dropped
ORB-SLAM3, GPU, LC on	0.52	0.44	1.11	0.057	0.85	2	2420
ORB-SLAM3, CPU, LC on	0.62	0.48	1.38	0.110	0.88	2	4363
ORB-SLAM3, GPU, LC off	**0.46**	0.36	0.99	0.056	0.87	0	3201
ORB-SLAM3, CPU, LC off	1.03	0.89	1.83	0.223	0.96	3	4204

Table 3.7: The same ORB-SLAM3 run (GPU, LC on) scored against the tracker export as delivered and with one sample removed (the 5.2 m excursion at 184.9 s). The time offsets are the values the wheel-speed cross-correlation found on each export on 2 November (peak correlation 0.10 raw, 0.61 cleaned), applied here as pinned values. The cleaned truth still fails the gap rule (66 gaps above 1 s, tracker at 1.7 Hz).

Truth	GT path [m]	offset [s]	ATE [m]	median [m]	path ratio
as delivered	36.8	4.38	1.174	0.851	0.65
one sample removed	27.1	-4.14	0.519	0.437	0.85

Analysis. The front end is what separates the runs. The GPU extractor drops a third to a half of the frames and holds 0.5 m; the CPU extractor drops two thirds and loses a factor of two, one of its runs with a 2 m vertical excursion the tool flags as implausible and the other with three resets and a half-second of lost tracking. Loop closing changes nothing that the front end does not change more. The remaining resets are an initialisation effect, two or three `local_mapping_bad_imu` map resets inside the first 13 s and none after (zero in the GPU run without loop closing); after them the error is a slow wave between 0.05

and 1 m, not an accumulating drift (figure 3.9), and it does not rise while turning in any run (table 3.8). The estimates are a uniform 12–15 % smaller than the tracker's path, which the unmeasured lever arm, a stereo-baseline error or a tracker scale error could each explain. One tracker sample decides the headline: scored against the export as delivered, the GPU run reads 1.17 m with a path ratio of 0.65, because that sample adds 10 m of false path and dominates the time alignment (table 3.7).

Table 3.8: ORB-SLAM3 on `dataset_realv6_000`, cleaned truth: growth rate of the planar error while the aligned estimate's heading changes faster than $10\,^\circ\,s^{-1}$ and on straights, with the time in each state. Positive = error growing. One run per configuration, so no test is possible.

Run	turning [mm/s]	straight [mm/s]	time turning [s]	time straight [s]
ORB-SLAM3, GPU, LC on	-11.2	-0.3	23	171
ORB-SLAM3, CPU, LC on	-29.8	-1.4	23	162
ORB-SLAM3, GPU, LC off	4.9	-2.1	11	185
ORB-SLAM3, CPU, LC off	-49.9	-0.9	20	182

The setting was reached by elimination. With the IMU noise densities copied from the VINS file the estimator was unstable on this recording, running away in its velocity state at a different place in each run; six out-of-bounds reads in the library were fixed without removing it, every other candidate was varied without removing it, and raising the IMU noise densities tenfold removed it in every run since. Those earlier runs were overwritten and carry no number here; the search is in the ORB-SLAM3 engineering record.

VINS-Fusion's failure is a delivery defect. The node received 3 187 of 6 601 frames but used 6 498 of the 43 494 IMU messages it received, reset 217 times on `imu_gap`, and stopped after 19 s; behind that defect the configuration still fixes a gravity norm of 9.2642 against the $9.878\,m\,s^{-2}$ this IMU reads at rest.

Table 3.9: VINS-Fusion on `dataset_realv6_000`, 3 November 2026 (`config/wil/stereo_imu.yaml`, IMU noise $\times 10$, gravity norm still 9.2642 against the $9.878\,m\,s^{-2}$ this IMU reads at rest). The scored span is the 18 s it tracked before its last reset and is not a result.

Quantity	Value
camera frames received (bag: 6 601)	3187
IMU messages received / samples used	43494 / 6498
`imu_gap` resets	217
poses written / span of poses	286 / 19 s
tracker samples matched	41 of 43 in span
ATE over the tracked span	0.18 m (not a result)

Second recording set, realv7. Three further recordings of 3 November 2026 (`dataset_realv7_000`, `_001`, `_002`: 21.6, 31.7 and 23.4 m over 97, 134 and 104 s) have tracker exports that pass every validity rule (5–6 Hz, no gap, no step). Each carries the same four ORB-SLAM3 configurations and one VINS-Fusion run. As written in the live log (table 3.10, figure 3.11), a plane fitted to the ORB-SLAM3 estimate lies within 10° of horizontal in three runs of twelve and at 88–90° in five, every `000` run among them, with the plane normal exactly a body axis in every run; scored as written, those nine runs are not scorable.

The per-frame diagnostics give the mechanism. The IMU initialised 3–4 s after the first pose. The upstream rule that resets a map with less than 2 cm between consecutive keyframes in the first ten seconds after the initialisation then discarded it, the robot being still at rest. The new map starts in the camera frame of the reset instant, and in it the re-initialisation was starved: upstream attempts it only with an empty keyframe queue, which a keyframe every 0.25 s and 250 ms of map-point creation per keyframe never allowed. The IMU re-initialised after 37–92 s in seven runs, never in two, and early (6.7 s) in the

one run that was level. Until then the estimate is stereo-only in a map whose vertical is the camera's y axis, and the late re-initialisation rotates the map under the already-written poses (the 1.9–6.8 m steps).

A scorer-side repair was written while the cause was being found: it rotates the pre-initialisation poses by the rotation measured at the initialisation frame and levels each map's fitted plane, both without using the truth. It is kept for the record and is not used for any result below, because the defect it worked around is now fixed in the estimator; the one figure it produced that is still worth quoting is that on `000` it recovered 0.252–0.275 m from runs whose estimates were lying on their side.

The fork's patch 15 adds two switches that attempt the initialisation with a queued backlog and skip the low-motion reset for stereo. On 11 November 2026 all twelve runs were re-made with both active, and table 3.11 reports them scored *raw*: neither of the two repairs above was applied, which is the point of the table.

The upstream behaviour is gone from every run. The IMU initialised at 2.8–4.7 s and was retained; thirteen of the sixteen request no map reset at all and the other three request one for a timestamp jump, not for insufficient motion. The plane of the estimate now lies 0.2–2.8° from horizontal in every run against the 5–90° of table 3.10, and the vertical range falls from as much as 9.6 to at most 1.00 m. Under the scorer's own rules seven `realv7` runs are now **completed** where none was before and the other five are **partial**; the four `realv6` runs remain rejected, not for anything the estimator did but because that recording's tracker export fails the validity rules.

The accuracy claim is narrower than the defect claim, and the three recordings disagree. Set against the same configuration under upstream initialisation *after both repairs* — each estimator at the best obtainable from it — patch 15 matches it on `realv7_000` (0.251–0.268 against 0.252–0.275 m), is mixed on `_001`, where two runs improve and two worsen and one of those reaches 1.278 against 0.817 m, and beats it on `_002` (0.172–0.198 against 0.472–2.115 m, every one of the four better than anything the repair obtained there) and on `realv6_000`, where all four improve (0.404–0.480 against 0.457–1.028 m) and the two CPU runs improve most. The defensible statement is therefore not that patch 15 is more accurate. It is that patch 15 removes the need for the repair: it reaches on the raw estimate what the stock estimator reaches only after its frame bookkeeping has been corrected offline, and on the one recording the repair handled worst it reaches considerably more. The 0.172–0.198 m runs of `_002` are the most accurate real-robot ORB-SLAM3 figures in this report.

[Open: what the improvement is owed to] Two things differ between the two sets of runs, not one. Patch 15 is the first; the second is that `IMU.NoiseGyro` and `IMU.NoiseAcc` were inflated tenfold on 3 November, and no artefact records which values that afternoon's runs used, because `run_metadata.csv` stores the configuration's path and not its contents. The loss of the tilt is patch 15's: the upstream rule tests keyframe spacing below 2 cm within 10 s and cannot be influenced by an inertial noise density. Any part of the accuracy difference is not separable on the present evidence. One control run, the same bag at today's noise with both switches returned to upstream, would separate them and has not been made.

[Provenance: the 3 November estimates are gone] The re-runs overwrote `output/output_orb/real/dataset_realv7_00*/` in place, so the pre-patch-15 estimates of all three recordings no longer exist and neither table 3.10 nor figure 3.11 can be regenerated; the frozen copies beside the generator are the only record of the behaviour this section reports as corrected. A run directory should be written under a name that carries its configuration, so that a re-run cannot overwrite the evidence for the behaviour it replaces.

VINS-Fusion diverges on all three with the IMU stream intact (zero resets) and 40–45 % of the frames received (table 3.12); its gravity norm has since been set to the measured 9.886 m s^{-2}.

[Sparse reference] The tracker ran at 1.7 Hz rather than its nominal 7 Hz on `realv6`, and the validity rules were written for the faster export. Either the export has to be re-captured at full rate or a rule for sparse references has to be written before any value from this recording is quoted outside a diagnostic.

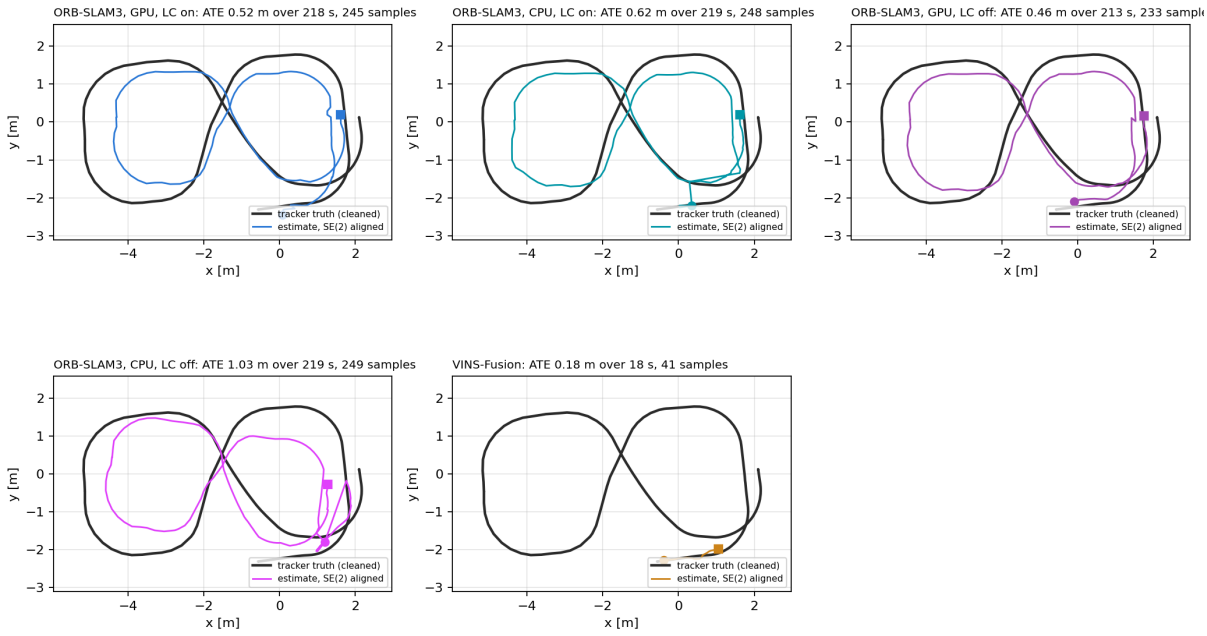


Figure 3.8: Every run against the cleaned tracker truth after rigid SE(2) alignment, one panel per run: ORB-SLAM3 with GPU or CPU extractor and loop closing on or off, and VINS-Fusion's 18 s.

Table 3.10: ORB-SLAM3 on the three `dataset_realv7` recordings, 3 November 2026 afternoon, same setting as the realv6 runs: the plane of each estimate as written in the live log, before any repair. "Tilt" is the angle between the plane fitted to the estimate's 3-D positions and the horizontal; "off-plane" the rms distance of the positions from that plane. A floor robot's estimate should be horizontal: a tilt near 90° means the poses were written while the map was not gravity-aligned (IMU not initialised; the vertical is then the camera's *y* axis). The scores are in the patched runs of table 3.11.

Recording	Run	tilt [°]	off-plane [m]	z range [m]	resets
`realv7_000`	GPU, LC on	89.1	0.35	6.87	3 s
`realv7_000`	GPU, LC off	89.5	0.02	6.98	3 s
`realv7_000`	CPU, LC on	88.8	0.41	6.88	5 s
`realv7_000`	CPU, LC off	87.9	0.58	6.79	4 s
`realv7_001`	GPU, LC on	88.8	0.05	9.64	4 s
`realv7_001`	GPU, LC off	6.1	1.12	5.77	3 s, 5 s
`realv7_001`	CPU, LC on	1.5	0.07	0.38	4 s
`realv7_001`	CPU, LC off	13.0	1.05	5.23	5 s
`realv7_002`	GPU, LC on	37.7	0.87	4.08	3 s, 49 s, 59 s
`realv7_002`	GPU, LC off	59.7	0.83	4.80	49 s
`realv7_002`	CPU, LC on	5.2	0.73	2.94	none
`realv7_002`	CPU, LC off	21.7	0.84	7.03	none

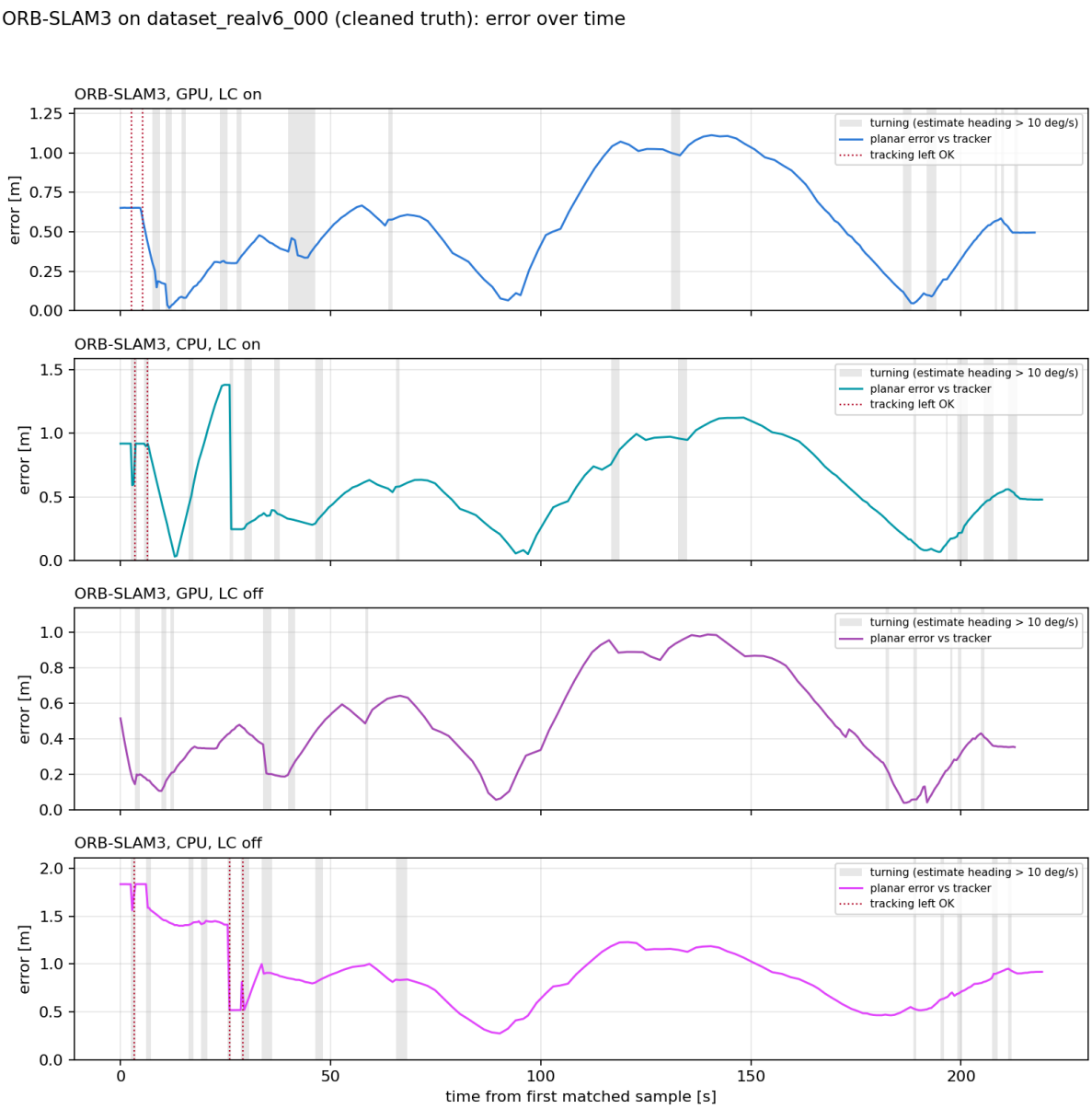


Figure 3.9: ORB-SLAM3 planar error over the run, one panel per configuration; grey bands are turns, dotted red the initialisation resets.

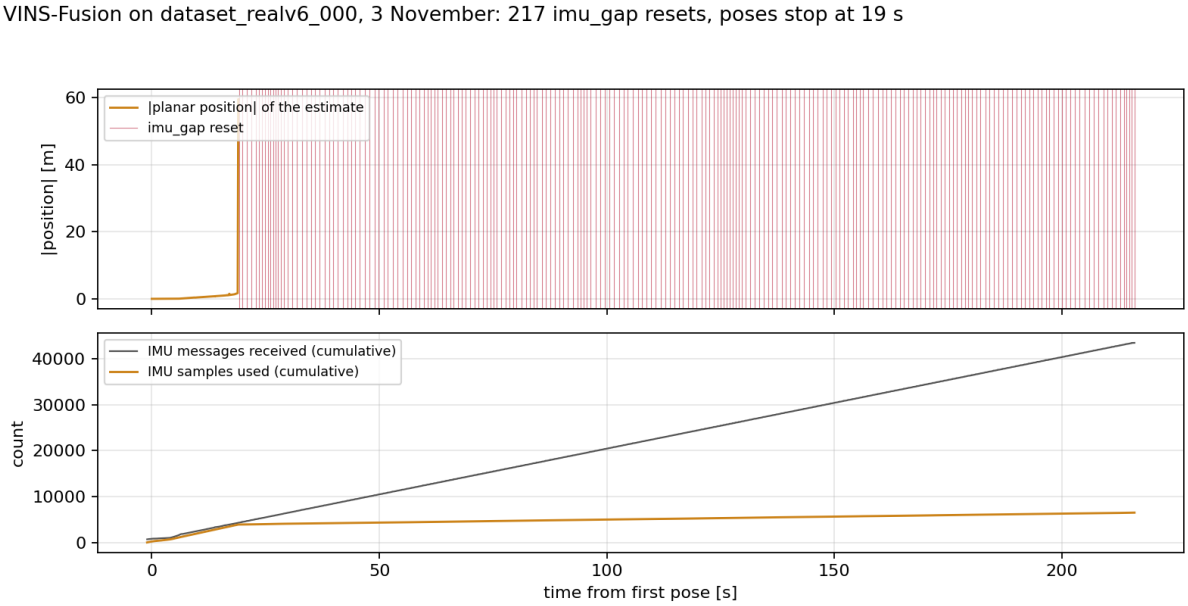


Figure 3.10: VINS-Fusion: distance from origin with every `imu_gap` reset marked, and IMU messages received against samples used.

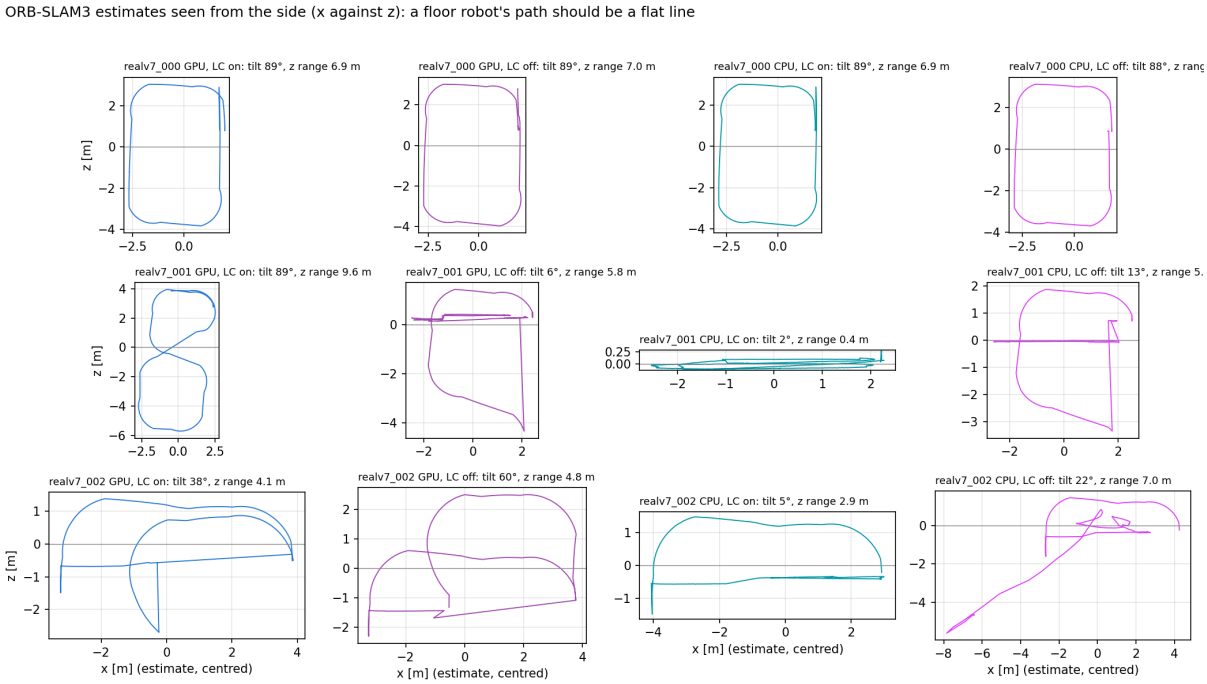


Figure 3.11: The twelve `realv7` ORB-SLAM3 estimates as written in the live log, seen from the side (x against z, centred); a floor robot's path should be a flat line. This is the defect that fork patch 15 removes; the estimates it produces instead are figure 3.12.

Table 3.11: Every real-robot ORB-SLAM3 run on the four recordings that carry an independent planar reference, with fork patch 15 active (`LocalMapping.resetOnLowMotion` 0, `LocalMapping.initWithBacklog` 1) and scored *raw*: neither `-join-imu-init` nor `-level-plane`, the repairs every other `realv7` figure in this chapter depends on. "Tilt" and "z range" are measured the same way as in table 3.10, so the two are comparable. "ATE before" is the same configuration on the same recording on 3 November, scored raw. Front-end identity was confirmed from the median tracking latency (GPU 41–50 ms, CPU 104–113 ms), not from the run name. "Stock + repair" is the same configuration on the same recording on 3 November, under upstream initialisation and, for `realv7`, only after the scorer-side frame repair that has since been retired: it is the fair comparison, each estimator at the best obtainable from it. For `realv6_000`, which never showed the tilt and needed no repair, "before" is the same run scored against the same cleaned tracker export without the switches. Status is the scorer's: it rejects a run whose estimate spans more than 1 m vertically, and it rejects every `realv6` run because that recording's tracker export is 1.7 Hz and fails the validity rules written for the 7 Hz one, so its rows are diagnostics. Lever arm still unmeasured, so every value is provisional.

Recording	Run	IMU init [s]	resets	tilt [°]	z range [m]	Kept	ATE [m]	Before [m]	status
`realv6_000`	GPU, LC on	3.1	0	1.1	0.27	0.58	0.478	0.519	failed
	GPU, LC off	4.5	1	1.2	0.33	0.51	0.404	0.457	failed
	CPU, LC on	3.0	0	0.2	0.31	0.30	0.480	0.618	failed
	CPU, LC off	2.8	0	1.8	0.36	0.28	0.456	1.028	failed
`realv7_000`	GPU, LC on	3.0	0	2.1	0.37	0.75	0.268	0.252	partial
	GPU, LC off	3.0	0	2.0	0.36	0.69	0.251	0.259	partial
	CPU, LC on	4.3	1	2.6	0.39	0.34	0.266	0.275	partial
	CPU, LC off	4.3	1	1.6	0.34	0.32	0.259	0.261	partial
`realv7_001`	GPU, LC on	2.9	0	2.0	0.42	0.71	0.451	0.477	completed
	GPU, LC off	4.7	0	1.7	0.45	0.67	1.278	0.817	completed
	CPU, LC on	2.8	0	0.8	0.40	0.33	0.575	0.438	completed
	CPU, LC off	2.9	0	2.0	0.42	0.32	0.442	0.558	completed
`realv7_002`	GPU, LC on	3.0	0	2.0	0.43	0.57	0.172	2.115	completed
	GPU, LC off	3.1	0	2.5	0.45	0.63	0.198	2.079	partial
	CPU, LC on	3.2	0	2.5	0.87	0.32	0.194	0.472	completed
	CPU, LC off	3.2	0	2.8	1.00	0.31	0.198	1.675	completed

Table 3.12: VINS-Fusion on the three `dataset_realv7` recordings (`config/wil/stereo_imu.yaml`, IMU noise $\times 10$, gravity norm 9.2642). Frames received against the bag's count (30 Hz over the recording), IMU use, resets, and where the estimate ended.

Recording	frames received	IMU received / used	resets	z range [m]	$\lvert p \rvert$ at end [m]
`realv7_000`	1155 of about 2663	17906 / 19053	0	12834403	37554
`realv7_001`	1541 of about 3686	24809 / 26340	0	3701	15974
`realv7_002`	1271 of about 2910	19368 / 20627	0	25937	5252

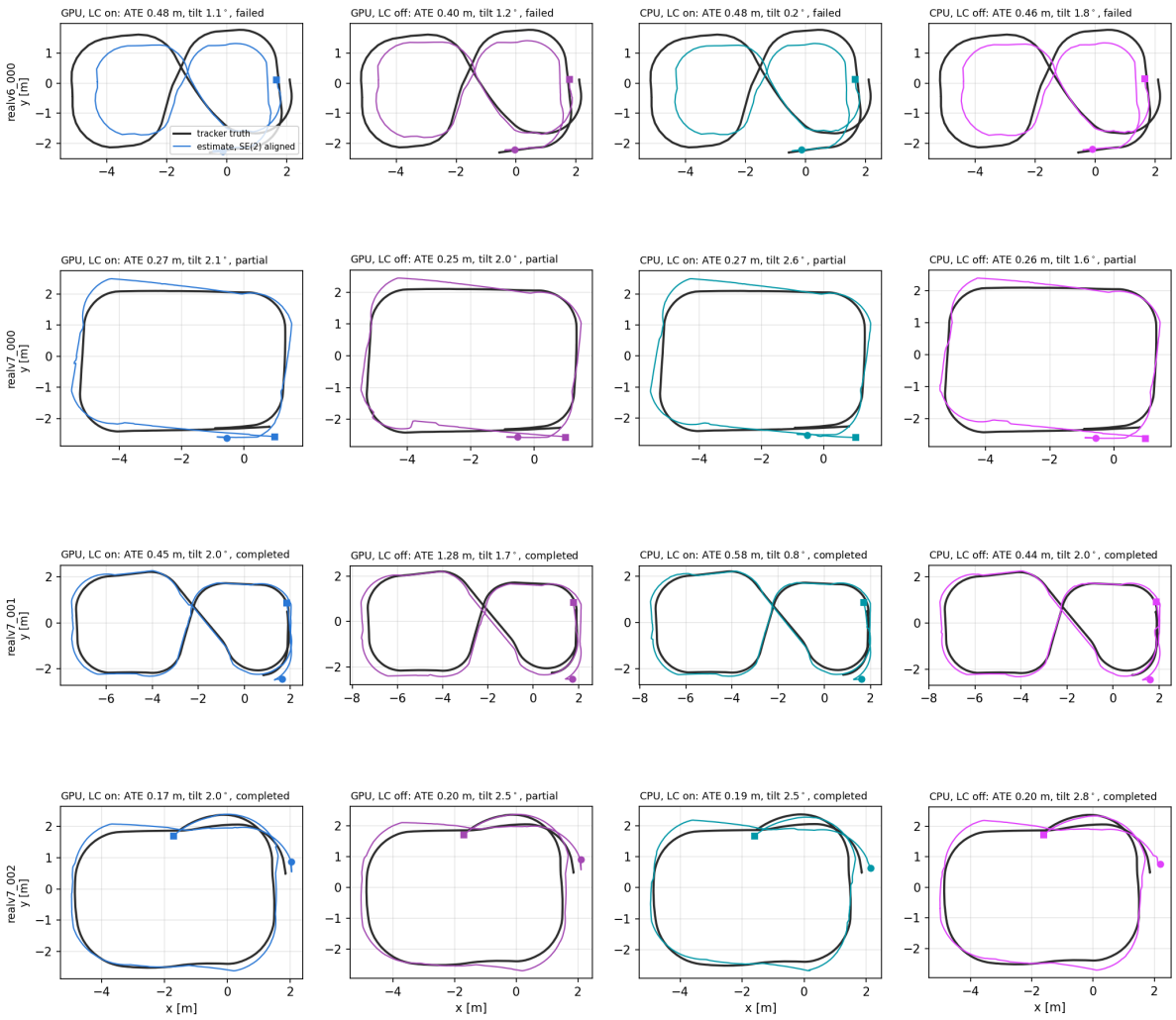


Figure 3.12: Every real-robot ORB-SLAM3 run on the four recordings that carry an independent planar reference, with fork patch 15 active, against the tracker truth after rigid SE(2) alignment. No repair of any kind is applied: these are the estimates as the node wrote them, where those of figure 3.11 had to be levelled and rotated before they could be drawn at all. Each panel gives the planar ATE, the tilt of the fitted plane and the scorer's status.

Summary. The real robot has an unfiltered ORB-SLAM3 reference at a provisional 0.46–0.52 m on `realv6` (GPU front end) and, on the three `realv7` recordings with valid truth, 0.251–0.268, 0.442–1.278 and 0.172–0.198 m with no repair applied at all. The cause of the IMU staying uninitialised for most of nine of the original twelve runs — an upstream map reset at the initialisation and a starved re-initialisation — is diagnosed, patched in the fork and now exercised: with the two switches active the reset does not occur in any of the twelve, every estimate is level, and the scorer's frame repair is no longer needed. What the patch is not shown to do is improve accuracy, which it matches on one recording, splits on a second and improves on the third. Also here: a measured cost of the CPU front end at real time, and two diagnosed VINS-Fusion defects, one of them (the gravity norm) now corrected in the configuration. The experiment's own question, whether filtering helps, is still untested on real imagery.

Next action. Run one `realv7` recording with both initialisation switches returned to upstream at the current inertial noise, which is the control that separates patch 15 from the tenfold noise change made the same week; re-run VINS-Fusion with the corrected gravity norm; measure the marker-to-IMU lever arm, which every real figure in this chapter is provisional on; then run both pipelines filtered and unfiltered on the same recording, with the detector live, and score them with the admissibility gates of the simulation slots. Write each run under a directory name that carries its configuration, so that no re-run overwrites the evidence it replaces.